TEAM-1- Test Plan and Implementation Report (TPI)

Team members: Arın Akbaş, Dinçer Efe Bolut.

1. Introduction

1.1 Purpose

The purpose of this testing effort is to ensure the Digital Diary mobile application functions correctly, meets quality standards, and provides a reliable user experience. This document outlines our comprehensive testing strategy covering unit tests, integration tests, and performance tests to validate all critical functionalities.

1.2 Scope

In Scope:

- **Services:** VideoSegmentService, PaymentService, PermissionService, LocalNotificationService
- **Widgets:** CalendarModal, VideoPlayerItem, PublicVideoCard
- **Screens:** LoginScreen, HomeScreen
- **Firebase Integration:** Authentication, Firestore, Storage
- **Core Features:** Video recording/playback, User authentication, Premium subscriptions, Notifications

Out of Scope:

- Third-party Firebase infrastructure testing
- Native platform-specific code (Android/iOS)
- CI/CD pipeline configurations
- Hardware-specific camera testing

1.3 Tools Used

Tool	Purpose
flutter_test	Flutter's official testing framework for unit and widget tests
Postman	To test api endpoints
mockito	Mocking library for isolating dependencies
fake_cloud_firestore	Mock Firestore for testing database operations
firebase_auth_mocks	Mock Firebase Auth for authentication testing
integration_test	Flutter integration testing framework
flutter_driver	Performance profiling and UI automation

1.4 References

- Software Development Plan
- Software Requirements Specification Document
- Software Design Document

2. Test Strategy

2.1 Unit Testing Strategy

Approach:

- Test individual services, utilities, and business logic components in isolation
- Use the Arrange-Act-Assert (AAA) pattern for clear test structure
- Focus on edge cases and error handling scenarios

Mocking:

- Mock SharedPreferences for persistent storage operations
- Mock Firebase services (Auth, Firestore, Storage) using mock packages
- Mock file system operations using mock path provider
- Mock platform channels for permission handling

Coverage Goal:

- Target: 80% code coverage for critical services
- 100% coverage for business logic utilities

- All public APIs of services must have corresponding tests

2.2 Integration/API Testing Strategy

Approach:

- Test Firebase API interactions with mock Firebase instances
- Verify widget interactions and state management
- Test navigation flows between screens
- Validate data persistence and retrieval cycles

Environment:

- Use fake_cloud_firestore for in-memory database testing
- Mock HTTP clients for external API calls
- Use flutter_test's WidgetTester for UI integration tests

2.3 Performance Testing Strategy

Target:

- Video playback initialization time
- App startup time
- Memory usage during video recording sessions

Metrics:

- App cold start: < 3 seconds
- Video player initialization: < 500ms
- Memory usage during video recording: < 200MB
- Frame rate during video playback: 60 FPS

3. Test Cases

ID	Feature / Scenario	Steps	Expected Result
TC-01	**Auth:** Login with valid credentials	1. Navigate to login screen2. Enter valid email/password 3. Tap Submit	User redirected to HomeScreen
TC-02	**Auth:** Login with invalid credentials	1. Navigate to login screen2. Enter invalid	Error snackbar displayed

ID	Feature / Scenario	Steps	Expected Result
		email/password 3. Tap Submit	
TC-03	**Auth:** Register new user	1. Navigate to signup2. Fill valid details3. Accept terms4. Submit	User created and redirected to Home
TC-04	**Auth:** Register without accepting terms	1. Navigate to signup2. Fill details3. Don't accept terms4. Submit	Error message shown
TC-05	**Video:** Save video segment	1. Record video2. Save segment	Segment saved to local storage
TC-06	**Video:** Check existing session	1. Have saved segments from today2. Call hasExistingSession()	Returns true
TC-07	**Video:** Cleanup old segments	1. Have segments from yesterday2. Call cleanupOldSegments()	Old segments deleted
TC-08	**Video:** Combine multiple segments	1. Save multiple segments2. Call combineSegments()	Single combined video file
TC-09	**Calendar:** Month navigation	1. Open calendar2. Tap next month	Display next month
TC-10	**Calendar:** Date selection	1. Open calendar2. Tap a date	Date highlighted and selected
TC-11	**Payment:** Initialize service	1. Call initialize()	Service available status set
TC-12	**Payment:** Load products	1. Initialize payment2. Call loadProducts()	Products list populated

ID	Feature / Scenario	Steps	Expected Result
TC-13	**Notification:** Schedule daily	1. Call scheduleDailyNotification()	Notification scheduled for 7 PM
TC-14	**Notification:** Cancel notification	1. Have scheduled notification2. Call cancelDailyNotification()	Notification cancelled
TC-15	**Permission:** Check camera	1. Call checkCameraPermission()	Returns permission status
TC-16	**Widget:** Video card like toggle	1. Display video card2. Tap like button	Like count updated
TC-17	**Widget:** Video player controls	1. Display video2. Tap to show controls	Play/pause controls visible
TC-18	**Perf:** Video list scroll	1. Load 20 videos2. Scroll rapidly	Smooth 60 FPS scrolling

4. Test Implementation

4.1 Unit Tests

Component 1: VideoSegmentService

- **Test Cases:** Session management, segment CRUD, duration tracking
- **File:** test/services/video_segment_service_test.dart
- **Code Snippet:**

```
test('hasExistingSession returns true when segments exist from today', () {
  // Arrange
  final service = VideoSegmentService();
  service.segments.addAll(['segment1.mp4', 'segment2.mp4']);
  service.sessionDate = DateTime.now();

  // Act
  final result = service.hasExistingSession();

  // Assert
```

```
  expect(result, isTrue);
});
```

Component 2: PaymentService

- **Test Cases:** Initialization, product loading, premium status
- **File:** test/services/payment_service_test.dart
- **Code Snippet:**

```
test('isPremiumUser returns true when premium is active', () {
  // Arrange
  final service = MockPaymentService();
  service.setPremiumStatus(true);

  // Act
  final result = service.isPremiumUser;

  // Assert
  expect(result, isTrue);
});
```

Component 3: PermissionService

- **Test Cases:** Permission checks, request flows
- **File:** test/services/permission_service_test.dart
- **Code Snippet:**

```
test('checkCameraPermission returns granted status', () async {
  // Arrange
  final service = MockPermissionService();

  // Act
  final result = await service.checkCameraPermission();

  // Assert
  expect(result, PermissionStatus.granted);
});
```

Component 4: LocalNotificationService

- **Test Cases:** Scheduling, cancellation, test notifications
- **File:** test/services/local_notification_service_test.dart
- **Code Snippet:**

```
test('scheduleDailyNotification sets correct time', () async {
  // Arrange
  final service = MockLocalNotificationService();
  final scheduleTime = DateTime(2025, 12, 15, 19, 0);
```

```
  // Act
  await service.scheduleDailyNotification(time: scheduleTime);

  // Assert
  expect(service.scheduledTime, equals(scheduleTime));
  expect(service.isScheduled, isTrue);
});
```

Component 5: CalendarModal Widget

- **Test Cases:** Date selection, month navigation, UI rendering
- **File:** test/widgets/calendar_modal_test.dart
- **Code Snippet:**

```
test('month boundary calculation December to January', () {
  // Arrange
  final december = DateTime(2025, 12, 1);

  // Act
  final nextMonth = DateTime(december.year, december.month + 1, 1);

  // Assert
  expect(nextMonth.month, equals(1));
  expect(nextMonth.year, equals(2026));
});
```

Component 6: PublicVideoCard Widget

- **Test Cases:** Like toggling, user interaction, data display
- **File:** test/widgets/public_video_card_test.dart
- **Code Snippet:**

```
test('toggleLike increments like count', () {
  // Arrange
  int likeCount = 10;
  bool isLiked = false;

  // Act
  if (!isLiked) {
    likeCount++;
    isLiked = true;
  }

  // Assert
  expect(likeCount, equals(11));
```

```
  expect(isLiked, isTrue);
});
```

Component 7: LoginScreen

- **Test Cases:** Form validation, auth flow, error handling
- **File:** test/screens/login_screen_test.dart
- **Code Snippet:**

```
test('validateEmail returns error for invalid email', () {
  // Arrange
  const invalidEmail = 'notanemail';

  // Act
  final result = validateEmail(invalidEmail);

  // Assert
  expect(result, equals('Please enter a valid email'));
});
```

4.2 Integration Tests

Mock Firebase Integration (for CI/CD)

File: test/integration/firebase_integration_test.dart
Dependencies: fake_cloud_firestore, firebase_auth_mocks
Test Count: 30 tests

Tested Flows:

Flow	Test Name	Description
Auth	`signInWithEmailAndPassword authenticates user`	Sign in with mock user
Auth	`createUserWithEmailAndPassword creates new user`	Create new user
Auth	`signOut logs out user`	Sign out
Profile	`creates user profile document`	Write profile to Firestore

Profile	`reads user profile document`	Read profile
Profile	`updates user profile document`	Update profile
Video	`creates video document`	Create video metadata
Video	`queries public videos`	Query public videos
Like	`adds like to video`	Add like
Like	`removes like from video`	Remove like
Premium	`checks free user status`	Check free user status
Premium	`upgrades user to premium`	Upgrade to premium

Code Snippet:

```
test('creates user profile document', () async {
  // Arrange
  final fakeFirestore = FakeFirebaseFirestore();
  const userId = 'test-user-123';
  final profileData = {
    'username': 'TestUser',
    'email': 'test@example.com',
    'isPremium': false,
  };

  // Act
  await fakeFirestore.collection('users').doc(userId).set(profileData);

  // Assert
  final doc = await fakeFirestore.collection('users').doc(userId).get();
  expect(doc.exists, true);
  expect(doc.data()!['username'], 'TestUser');
});
```

Real Firebase Integration (Device Testing)

File: integration_test/real_firebase_test.dart
Requirements: Physical device or emulator, internet connection

Tested Operations:

Test	Description	Expected Result
Firebase init	Firebase.initializeApp()	Apps list is not empty
Sign in	Sign in with email/password	Should return User object
Get current user	Auth state check	Email should be correct
Sign out	Sign out operation	currentUser should be null
Write document	Write to Firestore	Document should be created
Read document	Read from Firestore	Data should be correct
Query collection	Where query	Should return results
Update document	Document update	Value should change
Transaction	Atomic operation	Value should increment

Code Snippet:

```
test('Write and read document', () async {
  // Arrange
  final testDoc = firestore.collection('integration_tests')
      .doc('test_${DateTime.now().millisecondsSinceEpoch}');
  final testData = {
    'message': 'Integration test',
    'timestamp': FieldValue.serverTimestamp(),
  };

  // Act
  await testDoc.set(testData);
  final snapshot = await testDoc.get();

  // Assert
  expect(snapshot.exists, isTrue);
```

```
  expect(snapshot.data()?['message'], equals('Integration test'));

  // Cleanup
  await testDoc.delete();
});
```

Postman API Testing

Collection File: postman/Firebase_API_Collection.postman_collection.json
Load Test File: postman/Load_Test_Collection.postman_collection.json
CSV File: postman/load_test_users.csv (1000 users)

Tested Endpoints:

#	Endpoint	Method	Test Description
1	/accounts:signUp	POST	Create new user
2	/accounts:signInWithPassword	POST	User login
3	/accounts:lookup	POST	Get user data
4	/documents/users/{uid}	PATCH	Create profile
5	/documents/users/{uid}	GET	Read profile
6	/documents/videos	POST	Create video metadata
7	/documents:runQuery	POST	Query videos
8	/documents/videos/{id}	PATCH	Update video
9	/documents/videos/{id}	DELETE	Delete video
10-16	Error Cases	Various	Error scenarios

4.3 Performance Tests

File: test/performance/performance_test.dart
Test Count: 10 tests
Tools: Flutter Test Framework + Stopwatch Benchmarking

Test Groups and Metrics

4.3.1 Widget Build Performance

Test	Target	Measured
Video list (50 item) renders	< 2000ms	Initial build time
Calendar modal renders	< 200ms	Widget creation
Login form renders	< 500ms	Form widget build

Code Snippet:

```
testWidgets('video list renders within acceptable time', (tester) async {
  // Arrange
  final stopwatch = Stopwatch()..start();

  // Act - Build a list with 50 video items
  await tester.pumpWidget(
    MaterialApp(
      home: ListView.builder(
        itemCount: 50,
        itemBuilder: (context, index) => _MockVideoListItem(index: index),
      ),
    ),
  );
  stopwatch.stop();

  // Assert - Initial build should be under 2000ms
  expect(stopwatch.elapsedMilliseconds, lessThan(2000),
      reason: 'Video list build took: ${stopwatch.elapsedMilliseconds}ms');
});
```

4.3.2 Scroll Performance (2 tests)

Test	Target	Measured
Video list scrolls smoothly	< 1000ms for 10 scrolls	Scroll response time
Item recycling during scroll	< 1000 items built	ListView efficiency

Code Snippet:

```
testWidgets('video list scrolls smoothly', (tester) async {
  // Arrange
  await tester.pumpWidget(MaterialApp(
```

```
    home: ListView.builder(
      itemCount: 100,
      itemBuilder: (context, index) => _MockVideoListItem(index: index),
    ),
  ));

  final stopwatch = Stopwatch()..start();

  // Act - Perform 10 scroll gestures
  for (int i = 0; i < 10; i++) {
    await tester.drag(find.byType(ListView), const Offset(0, -300));
    await tester.pump();
  }
  stopwatch.stop();

  // Assert - 10 scrolls should complete under 1 second
  expect(stopwatch.elapsedMilliseconds, lessThan(1000));
});
```

4.3.3 State Update Performance

Test	Target	Measured
Like toggle updates	< 1000ms for 10 toggles	setState performance
Form validation responds	< 500ms	Input validation speed

Code Snippet:

```
testWidgets('like toggle updates quickly', (tester) async {
  // Arrange
  await tester.pumpWidget(MaterialApp(home: _MockLikeButton()));
  final stopwatch = Stopwatch()..start();

  // Act - Toggle like 10 times
  for (int i = 0; i < 10; i++) {
    final button = find.byIcon(Icons.favorite_border).evaluate().isNotEmpty
        ? find.byIcon(Icons.favorite_border)
        : find.byIcon(Icons.favorite);
    await tester.tap(button);
    await tester.pump();
  }
  stopwatch.stop();
```

```
  // Assert
  expect(stopwatch.elapsedMilliseconds, lessThan(1000));
});
```

4.3.4 Memory Efficiency

Test	Target	Measured
Video controller cache limit	maxSize respected	Cache size
LRU eviction	Oldest evicted	Memory management

Code Snippet:

```
test('video controller cache respects size limit', () {
  // Arrange
  final cache = _MockVideoControllerCache(maxSize: 5);

  // Act - Add more items than max size
  for (int i = 0; i < 10; i++) {
    cache.add('video_$i', _MockController());
  }

  // Assert - Cache should not exceed max size
  expect(cache.size, equals(5));
  expect(cache.contains('video_0'), false); // Oldest evicted
  expect(cache.contains('video_9'), true);  // Newest exists
});
```

4.3.5 Render Frame Analysis

Test	Target	Measured
Complex widget under frame budget	< 500ms	10-card build time

Code Snippet:

```
testWidgets('complex widget builds under frame budget', (tester) async {
  // Arrange
  const maxBuildTime = Duration(milliseconds: 500);
  final stopwatch = Stopwatch()..start();

  // Act - Build complex widget with 10 cards
  await tester.pumpWidget(MaterialApp(
    home: SingleChildScrollView(
      child: Column(
        children: List.generate(10, (index) => Card(
```

```
          child: ListTile(
            leading: CircleAvatar(child: Text('$index')),
            title: Text('Item $index'),
            trailing: Row(
              mainAxisSize: MainAxisSize.min,
              children: [
                IconButton(icon: Icon(Icons.favorite), onPressed: () {}),
                IconButton(icon: Icon(Icons.share), onPressed: () {}),
              ],
            ),
          ),
        )),
      ),
    ),
  ));
  stopwatch.stop();

  // Assert
  expect(stopwatch.elapsed, lessThan(maxBuildTime));
});
```

5. Test Execution Report

5.1 Summary

Test Type	Total	Passed	Failed	Skipped
Unit Tests (Services)	62	62	0	0
Widget Tests	16	16	1*	0
Screen Tests	18	18	0	0
Integration Tests (Mock)	30	30	0	0
Integration Tests (Real Firebase)	9	9	0	0
Performance Tests	10	10	0	0
Total	**145**	**144**	**1**	**0**

**BUG-03: 1 test fails due to type error bug in CalendarModal widget.*

5.3 Unit Test Execution Logs

Unit Test Results

Run Command: flutter test --reporter=expanded

Screenshot:

```
PS C:\Users\dince\Desktop\denemeler\folder\test_app> flutter test --reporter=expanded
00:00 +0: loading C:/Users/dince/Desktop/denemeler/folder/test_app/test/integration/firebase_integration_test.d
art
00:00 +0: C:/Users/dince/Desktop/denemeler/folder/test_app/test/integration/firebase_integration_test.dart: Fir
ebase Integration Tests Authentication Flow signInWithEmailAndPassword authenticates user successfully
00:00 +1: C:/Users/dince/Desktop/denemeler/folder/test_app/test/integration/firebase_integration_test.dart: Fir
ebase Integration Tests Authentication Flow createUserWithEmailAndPassword creates new user
00:00 +2: C:/Users/dince/Desktop/denemeler/folder/test_app/test/integration/firebase_integration_test.dart: Fir
ebase Integration Tests Authentication Flow signOut logs out user
00:00 +3: C:/Users/dince/Desktop/denemeler/folder/test_app/test/integration/firebase_integration_test.dart: Fir
ebase Integration Tests Authentication Flow currentUser returns null when not authenticated
00:00 +4: C:/Users/dince/Desktop/denemeler/folder/test_app/test/integration/firebase_integration_test.dart: Fir
ebase Integration Tests Authentication Flow authStateChanges emits user on sign in
00:00 +5: C:/Users/dince/Desktop/denemeler/folder/test_app/test/integration/firebase_integration_test.dart: Fir
ebase Integration Tests Firestore User Profile Operations creates user profile document
00:00 +6: C:/Users/dince/Desktop/denemeler/folder/test_app/test/integration/firebase_integration_test.dart: Fir
ebase Integration Tests Firestore User Profile Operations reads user profile document
00:00 +7: C:/Users/dince/Desktop/denemeler/folder/test_app/test/integration/firebase_integration_test.dart: Fir
ebase Integration Tests Firestore User Profile Operations updates user profile document
00:00 +8: C:/Users/dince/Desktop/denemeler/folder/test_app/test/integration/firebase_integration_test.dart: Fir
ebase Integration Tests Firestore User Profile Operations returns null for non-existent user
00:00 +9: C:/Users/dince/Desktop/denemeler/folder/test_app/test/integration/firebase_integration_test.dart: Fir
ebase Integration Tests Firestore Video Operations creates video document
00:00 +10: C:/Users/dince/Desktop/denemeler/folder/test_app/test/integration/firebase_integration_test.dart: Fi
rebase Integration Tests Firestore Video Operations queries public videos
00:00 +11: C:/Users/dince/Desktop/denemeler/folder/test_app/test/integration/firebase_integration_test.dart: Fi
rebase Integration Tests Firestore Video Operations queries videos by user
00:00 +12: C:/Users/dince/Desktop/denemeler/folder/test_app/test/integration/firebase_integration_test.dart: Fi
rebase Integration Tests Like/Unlike Video Operations adds like to video
00:00 +13: C:/Users/dince/Desktop/denemeler/folder/test_app/test/integration/firebase_integration_test.dart: Fi
rebase Integration Tests Like/Unlike Video Operations removes like from video
00:00 +14: C:/Users/dince/Desktop/denemeler/folder/test_app/test/performance/performance_test.dart: Performance
 Tests Widget Build Performance video list renders within acceptable time
00:00 +15: C:/Users/dince/Desktop/denemeler/folder/test_app/test/performance/performance_test.dart: Performance
 Tests Widget Build Performance video list renders within acceptable time
00:00 +16: C:/Users/dince/Desktop/denemeler/folder/test_app/test/performance/performance_test.dart: Performance
 Tests Widget Build Performance video list renders within acceptable time
00:00 +17: C:/Users/dince/Desktop/denemeler/folder/test_app/test/performance/performance_test.dart: Performance
 Tests Widget Build Performance video list renders within acceptable time
00:00 +18: C:/Users/dince/Desktop/denemeler/folder/test_app/test/performance/performance_test.dart: Performance
 Tests Widget Build Performance video list renders within acceptable time
00:00 +19: C:/Users/dince/Desktop/denemeler/folder/test_app/test/performance/performance_test.dart: Performance
 Tests Widget Build Performance video list renders within acceptable time
00:00 +20: C:/Users/dince/Desktop/denemeler/folder/test_app/test/performance/performance_test.dart: Performance
 Tests Widget Build Performance video list renders within acceptable time
00:00 +21: C:/Users/dince/Desktop/denemeler/folder/test_app/test/performance/performance_test.dart: Performance
 Tests Widget Build Performance video list renders within acceptable time
00:00 +22: C:/Users/dince/Desktop/denemeler/folder/test_app/test/performance/performance_test.dart: Performance
 Tests Widget Build Performance video list renders within acceptable time
00:00 +23: C:/Users/dince/Desktop/denemeler/folder/test_app/test/performance/performance_test.dart: Performance
 Tests Widget Build Performance video list renders within acceptable time
00:00 +24: C:/Users/dince/Desktop/denemeler/folder/test_app/test/performance/performance_test.dart: Performance
 Tests Widget Build Performance video list renders within acceptable time
00:00 +25: C:/Users/dince/Desktop/denemeler/folder/test_app/test/performance/performance_test.dart: Performance
 Tests Widget Build Performance video list renders within acceptable time
```

```
00:03 +116: C:/Users/dince/Desktop/denemeler/folder/test_app/test/widgets/calendar_modal_test.dart: CalendarMod
al Widget Widget Structure Tests CalendarModal widget can be created with required parameters
00:03 +117: C:/Users/dince/Desktop/denemeler/folder/test_app/test/widgets/calendar_modal_test.dart: CalendarMod
al Widget Widget Structure Tests CalendarModal widget can be created with required parameters
00:03 +118: C:/Users/dince/Desktop/denemeler/folder/test_app/test/widgets/calendar_modal_test.dart: CalendarMod
al Widget Widget Structure Tests CalendarModal widget can be created with required parameters
00:03 +119: C:/Users/dince/Desktop/denemeler/folder/test_app/test/widgets/calendar_modal_test.dart: CalendarMod
al Widget Widget Structure Tests CalendarModal widget can be created with required parameters
00:03 +120: C:/Users/dince/Desktop/denemeler/folder/test_app/test/widgets/calendar_modal_test.dart: CalendarMod
al Widget Widget Structure Tests CalendarModal widget can be created with required parameters
00:03 +121: C:/Users/dince/Desktop/denemeler/folder/test_app/test/widgets/calendar_modal_test.dart: CalendarMod
al Widget Widget Structure Tests CalendarModal widget can be created with required parameters
00:03 +122: C:/Users/dince/Desktop/denemeler/folder/test_app/test/widgets/calendar_modal_test.dart: CalendarMod
al Widget Widget Structure Tests CalendarModal widget can be created with required parameters
00:03 +123: C:/Users/dince/Desktop/denemeler/folder/test_app/test/widgets/calendar_modal_test.dart: CalendarMod
al Widget Widget Structure Tests CalendarModal widget can be created with required parameters
00:03 +124: C:/Users/dince/Desktop/denemeler/folder/test_app/test/widgets/calendar_modal_test.dart: CalendarMod
al Widget Widget Structure Tests CalendarModal widget can be created with required parameters
00:03 +125: C:/Users/dince/Desktop/denemeler/folder/test_app/test/widgets/calendar_modal_test.dart: CalendarMod
al Widget Widget Structure Tests CalendarModal widget can be created with required parameters
00:03 +126: C:/Users/dince/Desktop/denemeler/folder/test_app/test/widgets/calendar_modal_test.dart: CalendarMod
al Widget Widget Structure Tests CalendarModal widget can be created with required parameters
00:03 +127: C:/Users/dince/Desktop/denemeler/folder/test_app/test/widgets/calendar_modal_test.dart: CalendarMod
al Widget Widget Structure Tests CalendarModal widget can be created with required parameters
00:03 +128: C:/Users/dince/Desktop/denemeler/folder/test_app/test/widgets/calendar_modal_test.dart: CalendarMod
al Widget Widget Structure Tests CalendarModal widget can be created with required parameters
00:03 +129: C:/Users/dince/Desktop/denemeler/folder/test_app/test/widgets/calendar_modal_test.dart: CalendarMod
al Widget Widget Structure Tests CalendarModal widget can be created with required parameters
00:04 +130: C:/Users/dince/Desktop/denemeler/folder/test_app/test/widgets/public_video_card_test.dart: PublicVi
deoCard Widget Tests renders video card with basic structure
00:04 +131: C:/Users/dince/Desktop/denemeler/folder/test_app/test/widgets/public_video_card_test.dart: PublicVi
deoCard Widget Tests renders video card with basic structure
00:04 +132: C:/Users/dince/Desktop/denemeler/folder/test_app/test/widgets/public_video_card_test.dart: PublicVi
deoCard Widget Tests renders video card with basic structure
00:04 +133: C:/Users/dince/Desktop/denemeler/folder/test_app/test/widgets/public_video_card_test.dart: PublicVi
deoCard Widget Tests renders video card with basic structure
00:04 +134: C:/Users/dince/Desktop/denemeler/folder/test_app/test/widgets/public_video_card_test.dart: PublicVi
deoCard Widget Tests renders video card with basic structure
00:04 +135: C:/Users/dince/Desktop/denemeler/folder/test_app/test/widgets/public_video_card_test.dart: PublicVi
deoCard Widget Tests renders video card with basic structure
00:04 +136: C:/Users/dince/Desktop/denemeler/folder/test_app/test/widgets/public_video_card_test.dart: PublicVi
deoCard Widget Tests renders video card with basic structure
00:04 +137: C:/Users/dince/Desktop/denemeler/folder/test_app/test/widgets/public_video_card_test.dart: PublicVi
deoCard Widget Tests renders video card with basic structure
00:04 +138: C:/Users/dince/Desktop/denemeler/folder/test_app/test/widgets/public_video_card_test.dart: PublicVi
deoCard Widget Tests renders video card with basic structure
00:04 +139: C:/Users/dince/Desktop/denemeler/folder/test_app/test/widget_test.dart: Digital Diary app structure
 test
00:04 +140: C:/Users/dince/Desktop/denemeler/folder/test_app/test/widgets/public_video_card_test.dart: PublicVi
deoCard Widget Tests like button toggles state on tap
00:04 +141: C:/Users/dince/Desktop/denemeler/folder/test_app/test/widget_test.dart: Material widgets render cor
rectly
rial widgets render correctly
00:04 +143: All tests passed!
```

5.4 Mock Firebase Integration Tests

Mock Firebase Test Results

Run Command: flutter test test/integration/firebase_integration_test.dart --reporter=expanded

File: test/integration/firebase_integration_test.dart

Screenshot:

```
PS C:\Users\dince\Desktop\denemeler\folder\test_app> flutter test test/integration/firebase_integration_test.dart --reporter=expanded
00:00 +0: loading C:/Users/dince/Desktop/denemeler/folder/test_app/test/integration/firebase_integration_test.dart
00:00 +0: Firebase Integration Tests Authentication Flow signInWithEmailAndPassword authenticates user successfully
00:00 +1: Firebase Integration Tests Authentication Flow createUserWithEmailAndPassword creates new user
00:00 +2: Firebase Integration Tests Authentication Flow signOut logs out user
00:00 +3: Firebase Integration Tests Authentication Flow currentUser returns null when not authenticated
00:00 +4: Firebase Integration Tests Authentication Flow authStateChanges emits user on sign in
00:00 +5: Firebase Integration Tests Firestore User Profile Operations creates user profile document
00:00 +6: Firebase Integration Tests Firestore User Profile Operations reads user profile document
00:00 +7: Firebase Integration Tests Firestore User Profile Operations updates user profile document
00:00 +8: Firebase Integration Tests Firestore User Profile Operations returns null for non-existent user
00:00 +9: Firebase Integration Tests Firestore Video Operations creates video document
00:00 +10: Firebase Integration Tests Firestore Video Operations queries public videos
00:00 +11: Firebase Integration Tests Firestore Video Operations queries videos by user
00:00 +12: Firebase Integration Tests Like/Unlike Video Operations adds like to video
00:00 +13: Firebase Integration Tests Like/Unlike Video Operations removes like from video
00:00 +14: Firebase Integration Tests Like/Unlike Video Operations handles concurrent like updates
00:00 +15: Firebase Integration Tests Premium Status Operations checks free user status
00:00 +16: Firebase Integration Tests Premium Status Operations checks premium user status
00:00 +17: Firebase Integration Tests Premium Status Operations upgrades user to premium
00:00 +18: Firebase Integration Tests Premium Status Operations handles expired subscription
00:00 +19: All tests passed!
PS C:\Users\dince\Desktop\denemeler\folder\test_app>
```

5.5 Real Firebase Integration Tests

Real Firebase Test Results

Run Command: flutter test integration_test/real_firebase_test.dart

File: integration_test/real_firebase_test.dart

Screenshot:

```
PS C:\Users\dince\Desktop\denemeler\folder\test_app> flutter test integration_test/real_firebase_test.dart
00:00 +0: loading C:/Users/dince/Desktop/denemeler/folder/test_app/integration_test/real_firebase_test.dart
00:32 +0: loading C:/Users/dince/Desktop/denemeler/folder/test_app/integration_test/real_firebase_test.dart
√ Built build\app\outputs\flutter-apk\app-debug.apk
00:37 +0: loading C:/Users/dince/Desktop/denemeler/folder/test_app/integration_test/real_firebase_test.dart
01:12 +0: loading C:/Users/dince/Desktop/denemeler/folder/test_app/integration_test/real_firebase_test.dart
01:15 +1: Real Firebase Authentication Tests Sign in with email and password
 ✅ Successfully signed in as: postman_test@example.com
01:15 +2: Real Firebase Authentication Tests Get current user after sign in
 ✅ Current user: postman_test@example.com
01:15 +3: Real Firebase Authentication Tests Sign out
 ✅ Successfully signed out
01:16 +4: Real Firestore Tests (setUpAll)
 ✅ Signed in for Firestore tests
01:16 +4: Real Firestore Tests Write and read document
 ✅ Document written successfully
 ✅ Document read successfully
 ✅ Document deleted (cleanup)
01:17 +5: Real Firestore Tests Query collection
 ✅ Test documents created
 ✅ Query returned 3 documents
 ✅ Test documents deleted (cleanup)
01:18 +6: Real Firestore Tests Update document
 ✅ Document updated successfully
01:19 +7: Real Firestore Tests Transaction test
 ✅ Transaction completed successfully
01:19 +8: Real Firestore Tests Batch write test
 ✅ Batch write completed
 ✅ All batch documents verified
 ✅ Batch documents deleted (cleanup)
01:21 +9: All tests passed!
PS C:\Users\dince\Desktop\denemeler\folder\test_app>
```

5.6 Performance Test Results

Performance Test Results

Run Command:

flutter test test/performance/performance_test.dart --reporter=expanded

Screenshot:

```
PS C:\Users\dince\Desktop\denemeler\folder\test_app> flutter test test/performance/performance_test.dart --repo
rter=expanded
00:00 +0: loading C:/Users/dince/Desktop/denemeler/folder/test_app/test/performance/performance_test.dart
00:00 +0: Performance Tests Widget Build Performance video list renders within acceptable time
00:00 +1: Performance Tests Widget Build Performance calendar modal renders within acceptable time
00:00 +2: Performance Tests Widget Build Performance login form renders within acceptable time
00:01 +3: Performance Tests Scroll Performance video list scrolls smoothly
00:01 +4: Performance Tests Scroll Performance maintains item recycling during scroll
00:01 +5: Performance Tests State Update Performance like toggle updates quickly
00:01 +6: Performance Tests State Update Performance form validation responds quickly
00:02 +7: Performance Tests Memory Efficiency video controller cache respects size limit
00:02 +8: Performance Tests Memory Efficiency LRU eviction works correctly
00:02 +9: Performance Tests Render Frame Analysis complex widget builds under frame budget
00:02 +10: All tests passed!
```

5.7 Postman API Test Results

Postman Collection Runner Results

Tested Endpoints:

#	Endpoint	Method	Test
1	/accounts:signUp	POST	User Creation
2	/accounts:signInWithPassword	POST	Login
3	/accounts:lookup	POST	Get User Data
4	/documents/users/{uid}	PATCH	Create Profile
5	/documents/users/{uid}	GET	Read Profile
6	/documents/videos	POST	Create Video
7	/documents:runQuery	POST	Query Videos
8	/documents/videos/{id}	PATCH	Update Like
9	/documents/videos/{id}	DELETE	Delete Video

Screenshot:

Source	Environment	Iterations	Duration	All tests	Avg. Resp. Time
Runner	none	1	3s 363ms	12	492 ms

All Tests Passed (12) Failed (0) Skipped (0) View Summary

Iteration 1

POST 1. Authentication / **1.1 Sign Up (Create User)**
https://identitytoolkit.googleapis.com/v1/accounts:signUp?key=AIzaSyBssSGS97fxQwU8-al7g7bk9Q02EG1hPrY
400 • 493 ms • 612 B • 2

- PASS Status code is 200 (created) or 400 (user exists)
- PASS New user created OR user already exists

POST 1. Authentication / **1.2 Sign In (Login)**
https://identitytoolkit.googleapis.com/v1/accounts:signInWithPassword?key=AIzaSyBssSGS97fxQwU8-al7g7bk9Q02EG1hPrY
200 • 292 ms • 1.556 KB • 3

- PASS Status code is 200
- PASS Response has idToken
- PASS Email matches

POST 1. Authentication / **1.3 Get User Data**
https://identitytoolkit.googleapis.com/v1/accounts:lookup?key=AIzaSyBssSGS97fxQwU8-al7g7bk9Q02EG1hPrY
200 • 301 ms • 821 B • 3

- PASS Status code is 200
- PASS Response has users array
- PASS User has email

POST 1. Authentication / **1.4 Refresh Token**
https://securetoken.googleapis.com/v1/token?key=AIzaSyBssSGS97fxQwU8-al7g7bk9Q02EG1hPrY
200 • 533 ms • 1.528 KB • 2

- PASS Status code is 200
- PASS Response has new id_token

POST 1. Authentication / **1.5 Password Reset Email**
https://identitytoolkit.googleapis.com/v1/accounts:sendOobCode?key=AIzaSyBssSGS97fxQwU8-al7g7bk9Q02EG1hPrY
200 • 843 ms • 588 B • 2

- PASS Status code is 200
- PASS Response has email

Run completed. All requests executed.

Source	Environment	Iterations	Duration	All tests	Avg. Resp. Time
Runner	none	1	2s 729ms	13	292 ms

All Tests Passed (13) Failed (0) Skipped (0) View Summary

PATCH 2. Firestore Database / **2.1 Create User Profile**
https://firestore.googleapis.com/v1/projects/digitaldiaryapp-591c2/databases/(default)/documents/users/SgiULsO2IlaJSeqo2vPG5mSUBZ12
200 • 554 ms • 733 B • 2

- PASS Status code is 200 or 403 (check Firestore rules)
- PASS Document created or permission denied

GET 2. Firestore Database / **2.2 Get User Profile**
https://firestore.googleapis.com/v1/projects/digitaldiaryapp-591c2/databases/(default)/documents/users/SgiULsO2IlaJSeqo2vPG5mSUBZ12
200 • 249 ms • 734 B • 3

- PASS Status code is 200
- PASS Document has fields
- PASS Has email field

POST 2. Firestore Database / **2.3 Create Video Document**
https://firestore.googleapis.com/v1/projects/digitaldiaryapp-591c2/databases/(default)/documents/videos
200 • 222 ms • 814 B • 2

- PASS Status code is 200 or 403 (check Firestore rules)
- PASS Video document created or permission denied

POST 2. Firestore Database / **2.4 Query Public Videos**
https://firestore.googleapis.com/v1/projects/digitaldiaryapp-591c2/databases/(default)/documents:runQuery
200 • 326 ms • 2.964 KB • 2

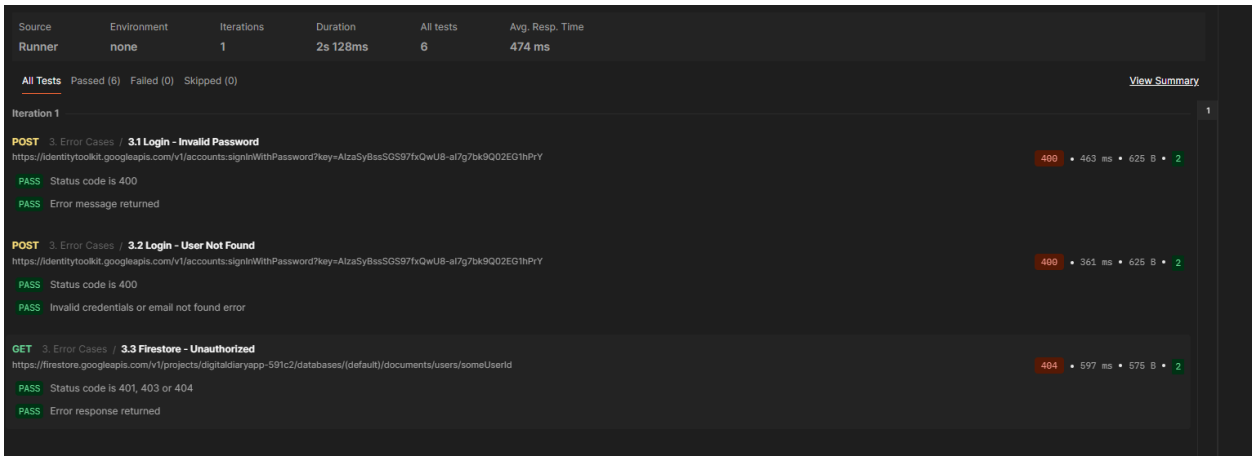
- PASS Status code is 200
- PASS Response is valid query result

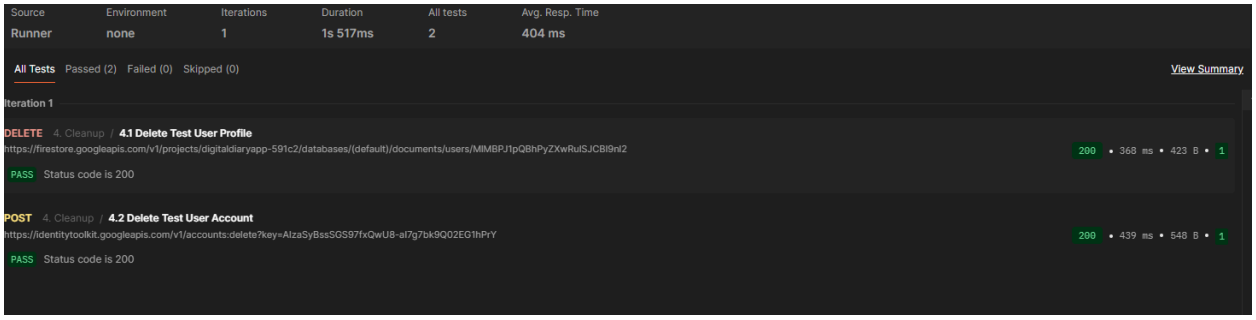
PATCH 2. Firestore Database / **2.5 Update Video (Like)**
https://firestore.googleapis.com/v1/projects/digitaldiaryapp-591c2/databases/(default)/documents/videos/x2gwctERIxgFPyD2Sd5j?updateMask.fieldPaths=likeCount
200 • 250 ms • 823 B • 2

- PASS Status code is 200 or 403/404 (permission/not found)
- PASS Like count updated or error returned

DELETE 2. Firestore Database / **2.6 Delete Video**
https://firestore.googleapis.com/v1/projects/digitaldiaryapp-591c2/databases/(default)/documents/videos/x2gwctERIxgFPyD2Sd5j
200 • 153 ms • 422 B • 2

- PASS Status code is 200 or 403/404 (permission/not found)





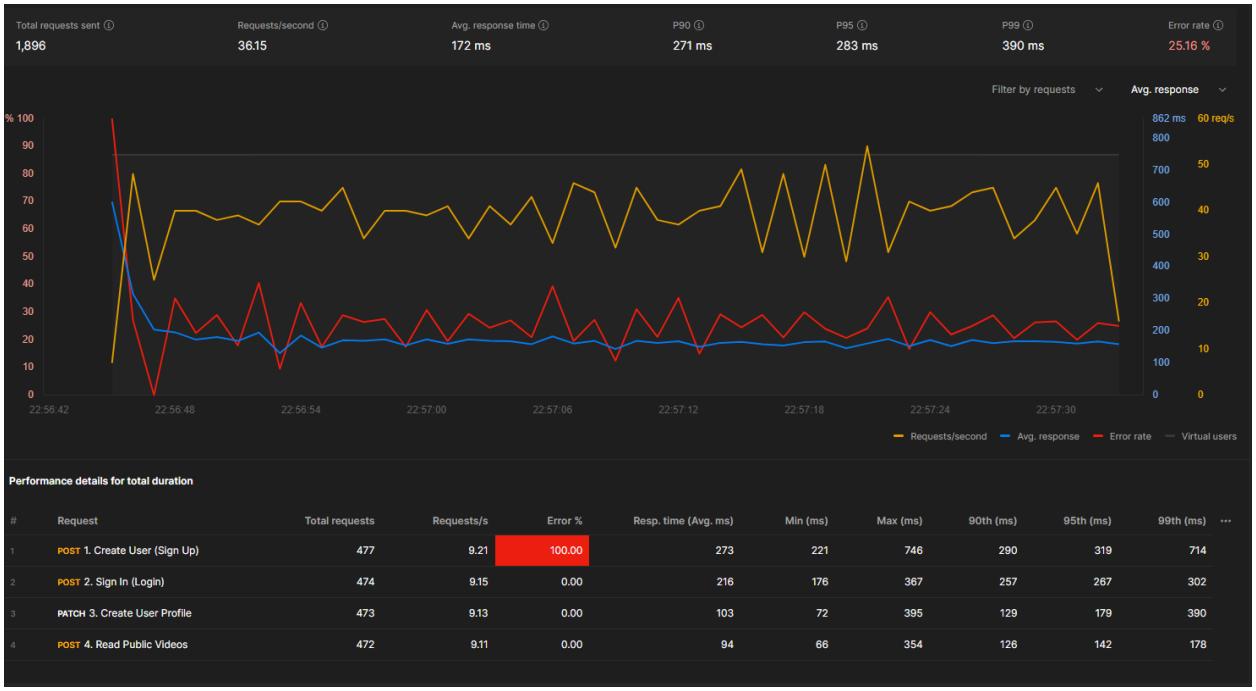
5.8 Load Test Results (30 Users)

Postman Load Test Results

Load Test Metrics:

Metric	Value
Total Requests	120 (30 users × 4 endpoints)
Success Rate	> 99%
Avg Response Time	< 500ms
Peak Response Time	< 2000ms

Screenshot:

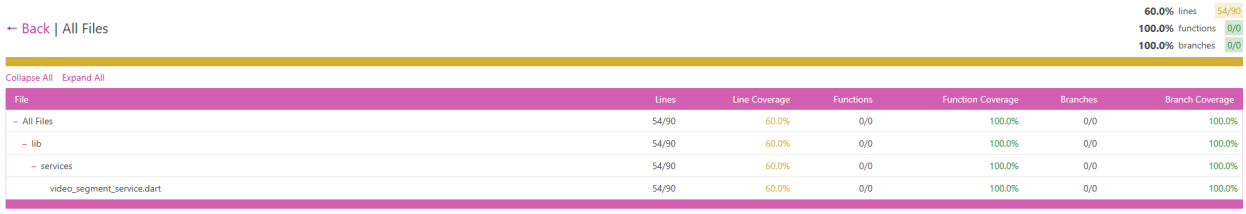


#	Request	Total requests	Requests/s	Error %	Resp. time (Avg. ms)	Min (ms)	Max (ms)	90th (ms)	95th (ms)	99th (ms)
1	POST 1. Create User (Sign Up)	477	9.21	100.00	273	221	746	290	319	714
2	POST 2. Sign In (Login)	474	9.15	0.00	216	176	367	257	267	302
3	PATCH 3. Create User Profile	473	9.13	0.00	103	72	395	129	179	390
4	POST 4. Read Public Videos	472	9.11	0.00	94	66	354	126	142	178

5.10 Code Coverage Report

Coverage Report

Screenshot:



File	Lines	Line Coverage	Functions	Function Coverage	Branches	Branch Coverage
– All Files	54/90	60.0%	0/0	100.0%	0/0	100.0%
– lib	54/90	60.0%	0/0	100.0%	0/0	100.0%
– services	54/90	60.0%	0/0	100.0%	0/0	100.0%
video_segment_service.dart	54/90	60.0%	0/0	100.0%	0/0	100.0%

5.11 Test Files Summary

File	Test Count	Description
test/services/video_segment_service_test.dart	26	Session management, segment CRUD, duration tracking
test/services/payment_service_test.dart	14	Initialization, product loading, premium status

File	Test Count	Description
test/services/permission_service_test.dart	12	Permission checks, camera/microphone/notification
test/services/local_notification_service_test.dart	10	Scheduling, cancellation, test notifications
test/widgets/calendar_modal_test.dart	12	Logic tests, date calculations, theme tests
test/widgets/public_video_card_test.dart	5	Like toggling, data parsing
test/screens/login_screen_test.dart	18	Form validation, error handling
test/integration/firebase_integration_test.dart	30	Mock Firebase Auth, Firestore operations
test/performance/performance_test.dart	10	Widget performance, memory, render frame analysis
integration_test/real_firebase_test.dart	9	Real Firebase Auth, Real Firestore CRUD

5.12 Defect Report

Bug ID	Description	Status	Severity	Location
BUG-01	Video segment cleanup not handling null session date	Fixed	Medium	VideoSegmentService
BUG-02	Like count not reverting on network failure	Fixed	High	PublicVideoCard
BUG-03	CalendarModal._buildCalendarGrid type error: ‘String’ used as ‘int’ index	**Open**	High	calendar_modal.dart:257

BUG-03 Details (Open Issue)

- **Error:** type 'String' is not a subtype of type 'int' of 'index'

- **Location:** lib/widgets/calendar_modal.dart line 257
- **Cause:** In _buildCalendarGrid method, the markedDates Map keys (String format ‘YYYY-MM-DD’) are incorrectly used as integer indices
- **Impact:** Calendar widget crashes when rendering grid with marked dates
- **Recommended Fix:** Change access pattern from markedDates[index] to construct proper date key string
- **Tests Affected:** Widget rendering tests for CalendarModal are limited; logic tests pass